retail

Invoice-line events without an expected purchase cadence. Source-local minute labels, cancellations, negative quantities and missing customer keys remain visible.

Completed evidence · 1 sources · 20 retained records

2 evidence-qualified finding(s) across 1 supplied source(s). Required stages completed over eligible records; behavioral search scope is disclosed separately.

Actions to investigate

> **Window effects meet configured investigation settings**
>
> Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.
>
> investigation_candidate · t8 · Quantity

> **Window effects meet configured investigation settings**
>
> Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.
>
> investigation_candidate · t12 · UnitPrice

Interpretation boundary

- No composite score or universal readiness verdict.
- Investigation defaults are not validated defect criteria.
- Supplied evidence does not prove what a production pipeline consumed.

Generated 2026-10-10T21:28:01.243477+00:00 · Schema 1.4 · temporal_reliability_v1

Declarations

Evidence / source	Measurement or rule	Retained result
retail	Source and base task	{"format":"csv","role":"dataset","task":"dataset_profile","targets":null}
retail	observation	event
retail	layout	wide
retail	entity columns	["CustomerID"]
retail	observation keys	[]
retail	clock	{"column":"time","kind":"local_datetime","encoding":"iso","format":null,"epoch_unit":null,"timezone":null,"timezone_column":null,"ambiguity":"unresolved","fold_column":null,"date_convention":null}
retail	schedule	{"kind":"none","cadence_ns":null,"anchor":null,"civil_unit":null,"timezone":null,"month_end":"skip","source_id":null,"time_column":null,"entity_columns":[],"measurement_column":null,"expectation":"required_record","collection_status_column":null,"collection_active_values":["active"]}
retail	active intervals	[]
retail	Quantity	{"name":"Quantity","column":null,"representation":"quantitative","unit":"items","meaning":null,"past_observation":false}
retail	UnitPrice	{"name":"UnitPrice","column":null,"representation":"quantitative","unit":"GBP","meaning":null,"past_observation":false}
retail	Country	{"name":"Country","column":null,"representation":"categorical","unit":null,"meaning":null,"past_observation":false}

11 of 11 metric rows shown. Presentation limits do not change analytical coverage.

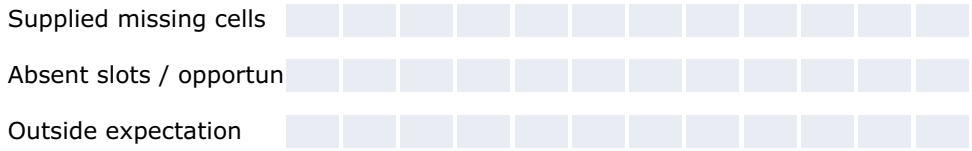
Chronology

Evidence / source	Measurement or rule	Retained result
t1 retail	rows	541909
t1 retail	cells	1625727
t1 retail	declared dense missing cells	0
t1 retail	parse states / valid	1625727
t1 retail	precision counts / second	1625727
t1 retail	value states [Country] / measurement	Country
t1 retail	value states [Country] / state	valid
t1 retail	value states [Country] / cells	541909
t1 retail	value states [Quantity] / measurement	Quantity
t1 retail	value states [Quantity] / state	valid
t1 retail	value states [Quantity] / cells	541909
t1 retail	value states [UnitPrice] / measurement	UnitPrice
t1 retail	value states [UnitPrice] / state	valid
t1 retail	value states [UnitPrice] / cells	541909
t1 retail	missing entity cells	405240
t1 retail	incomplete observation keys	0
t1 retail	observation meaning	event
t1 retail	clock units	nanoseconds
t1 retail	exact ticks	Yes
t3 retail	adjacent backward cells	0

20 of 30 metric rows shown. 10 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Continuity and history

Absent observations, supplied missing values and outside-expectation periods have different meanings. Event absence alone does not prove an outage.



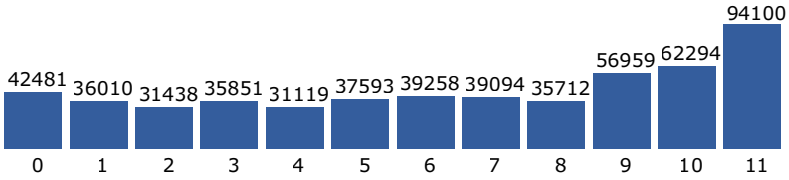
retail · missing values and absent slots remain separate

Evidence / source	Measurement or rule	Retained result
t5 retail	distinct entities	4372
t5 retail	distinct series	13116
t5 retail	history counts / min	1
t5 retail	history counts / max	7983
t5 retail	history counts / entity weighted mean	93.05329
t5 retail	history length bins [0] / observations	1
t5 retail	history length bins [0] / series	237
t5 retail	history length bins [1] / observations	2
t5 retail	history length bins [1] / series	1536
t5 retail	history length bins [2] / observations	3
t5 retail	history length bins [2] / series	7998
t5 retail	history length bins [3] / observations	4
t5 retail	history length bins [3] / series	3345
t5 retail	observed first tick	1291191960000000000
t5 retail	observed last tick	1323435000000000000
t5 retail	observed entry exit is lifecycle	No
t5 retail	retained histories [Country] / entity	["12503"]
t5 retail	retained histories [Country] / measurement	Country
t5 retail	retained histories [Country] / observations	1
t5 retail	retained histories [Country] / cells	1

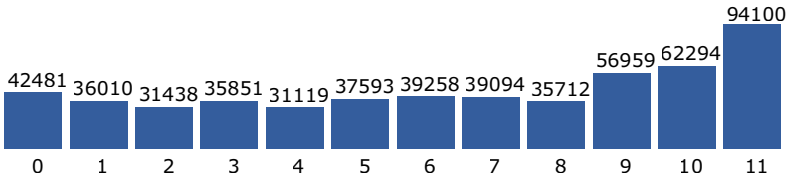
20 of 842 metric rows shown. 822 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Behavioral investigation

Thresholds and support settings are configurable investigation defaults. Reference support, sparse windows, zero dispersion, phase exclusions and population mix qualify interpretation.



retail / Quantity · supplied cells per window



retail / UnitPrice · supplied cells per window

Evidence / source	Measurement or rule	Retained result
t8 retail / Quantity	Reference and interpretation	{"reference_window":0,"unit":"items","meaning":"Observed effects are descriptive investigation evidence, not validated universal defects, changepoints, concept drift, or model performance."}
t8 retail / Quantity	Window 0	{"cells":42481,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":42481,"entities":949,"iqr":6.0}
t8 retail / Quantity	Window 1	{"cells":36010,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":36010,"entities":807,"iqr":8.0}
t8 retail / Quantity	Window 2	{"cells":31438,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":31438,"entities":891,"iqr":10.0}
t8 retail / Quantity	Window 3	{"cells":35851,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":35851,"entities":985,"iqr":9.0}
t8 retail / Quantity	Window 4	{"cells":31119,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":31119,"entities":942,"iqr":11.0}
t8 retail / Quantity	Window 5	{"cells":37593,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":37593,"entities":1070,"iqr":11.0}
t8 retail / Quantity	Window 6	{"cells":39258,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":39258,"entities":1058,"iqr":9.0}
t8 retail / Quantity	Window 7	{"cells":39094,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":39094,"entities":1017,"iqr":11.0}
t8 retail / Quantity	Window 8	{"cells":35712,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":35712,"entities":999,"iqr":11.0}
t8 retail / Quantity	Window 9	{"cells":56959,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":56959,"entities":1401,"iqr":10.0}
t8 retail / Quantity	Window 10	{"cells":62294,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":62294,"entities":1467,"iqr":11.0}
t8 retail / Quantity	Window 11	{"cells":94100,"missing_rate":0.0,"invalid":0,"nonfinite":0,"finite":94100,"entities":1757,"iqr":9.0}
t8 retail / Quantity	Comparison 0 → 1	{"reference_support":42481,"comparison_support":36010,"reference_iqr":6.0,"missingness_change":0.0,"cdf_distance":0.05317868194939 5123,"wasserstein_1":4.930342559140201,"normalized_w1":0.82172 37598567002,"distribution_review":true,"normalization_block":null}
t8 retail / Quantity	Comparison 1 → 2	{"reference_support":36010,"comparison_support":31438,"reference_iqr":8.0,"missingness_change":0.0,"cdf_distance":0.04351828883689 546,"wasserstein_1":5.597527625151802,"normalized_w1":0.699690 9531439752,"distribution_review":true,"normalization_block":null}
t8 retail / Quantity	Comparison 2 → 3	{"reference_support":31438,"comparison_support":35851,"reference_iqr":10.0,"missingness_change":0.0,"cdf_distance":0.0194693546363 63345,"wasserstein_1":1.2543447840961415,"normalized_w1":0.125 43447840961414,"distribution_review":false,"normalization_block":null}

Evidence / source	Measurement or rule	Retained result
t8 retail / Quantity	Comparison 3 → 4	{"reference_support":35851,"comparison_support":31119,"reference_iqr":9.0,"missingness_change":0.0,"cdf_distance":0.03223366381045656,"wasserstein_1":1.1338960934614106,"normalized_w1":0.12598845482904564,"distribution_review":false,"normalization_block":null}
t8 retail / Quantity	Comparison 4 → 5	{"reference_support":31119,"comparison_support":37593,"reference_iqr":11.0,"missingness_change":0.0,"cdf_distance":0.01736846770207512,"wasserstein_1":1.078976220338652,"normalized_w1":0.09808874730351382,"distribution_review":false,"normalization_block":null}
t8 retail / Quantity	Comparison 5 → 6	{"reference_support":37593,"comparison_support":39258,"reference_iqr":11.0,"missingness_change":0.0,"cdf_distance":0.03342412330487793,"wasserstein_1":1.695706815826716,"normalized_w1":0.154155165075156,"distribution_review":false,"normalization_block":null}
t8 retail / Quantity	Comparison 6 → 7	{"reference_support":39258,"comparison_support":39094,"reference_iqr":9.0,"missingness_change":0.0,"cdf_distance":0.03330010555996887,"wasserstein_1":1.8251334613451573,"normalized_w1":0.2027926068161286,"distribution_review":false,"normalization_block":null}

20 of 1839 metric rows shown. 1819 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 6 / 8

Availability and outcome maturity

Undeclared availability remains unknown. Historical agreement concerns supplied coverage; it does not establish what a pipeline consumed.

Evidence / source	Measurement or rule	Retained result
t7 retail	features	["Quantity","UnitPrice","Country"]
t7 retail	meaning	Undeclared availability remains unknown; no timestamp inheritance.

2 of 2 metric rows shown. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 7 / 8

Chronological evaluation

No retained evidence applies to this topic in the declared scope.

0 of 0 metric rows shown. Presentation limits do not change analytical coverage.

Methodology and coverage

Evidence / source	Measurement or rule	Retained result
method	schema	1.4
method	methodology	temporal_reliability_v1
method	execution status	completed
method	aggregate scope	Every eligible declared record in required stages; no silent sampling.
method	search scope [0] / selected measurements	["Quantity","UnitPrice","Country"]
method	search scope [0] / unsearched scope	[]
method	search scope [0] / work observation visits	13517082
method	search scope [0] / comparisons	525
method	search scope [0] / selection	Requested measurements first, then declaration order. Detail retention can miss behavior-only exceptions.
method	search scope [0] / defaults validated universal defect criteria	No
method	internal detail omissions / retail:entity histories	13016
method	internal detail omissions / retail:series effects	236
method	timezone / package	tzdata
method	timezone / version	2026.5
method	phase seconds / retail:ingestion	3.201941
method	phase seconds / retail:normalization	2.605882
method	phase seconds / required diagnostics	0.862993
method	phase seconds / availability history evaluation	1.670801e-05
method	phase seconds / behavior	2.426426
method	provisional limits / max cells	100000000

20 of 35 metric rows shown. 15 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Decoded-batch controls bound accepted batches and application retention. Native pages, dictionaries and conversion buffers can allocate before checks. Estimates and workload limits are provisional engineering controls. RSS monitoring can overshoot between samples.

Analytical scope left unsearched, retained-detail omissions, and presentation omissions are separate. No independent-row significance testing, model fitting, split generation, automatic repair or universal temporal score is performed.

Retained witnesses and selected detail

100 of 215 retained detail rows available here. Print shows up to 20 rows; JSON preserves every retained record. These presentation omissions do not change searched scope.

Evidence	Detail	Witness / selected population
t5	retained histories	{"entity":["12503"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1294314660000000000","last_tick":"1294314660000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12503"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1294314660000000000","last_tick":"1294314660000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12503"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1294314660000000000","last_tick":"1294314660000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12505"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1297431540000000000","last_tick":"1297431540000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12505"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1297431540000000000","last_tick":"1297431540000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12505"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1297431540000000000","last_tick":"1297431540000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12943"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1297437180000000000","last_tick":"1297437180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12943"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1297437180000000000","last_tick":"1297437180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["12943"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1297437180000000000","last_tick":"1297437180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13017"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1322841180000000000","last_tick":"1322841180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13017"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1322841180000000000","last_tick":"1322841180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13017"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1322841180000000000","last_tick":"1322841180000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13099"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1314898800000000000","last_tick":"1314898800000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13099"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1314898800000000000","last_tick":"1314898800000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13099"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1314898800000000000","last_tick":"1314898800000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13120"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1302862740000000000","last_tick":"1302862740000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13120"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1302862740000000000","last_tick":"1302862740000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13120"],"measurement":"UnitPrice","observations":1,"cells":1,"first_tick":"1302862740000000000","last_tick":"1302862740000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13135"],"measurement":"Country","observations":1,"cells":1,"first_tick":"1306493520000000000","last_tick":"1306493520000000000","span_ticks":"0"}
t5	retained histories	{"entity":["13135"],"measurement":"Quantity","observations":1,"cells":1,"first_tick":"1306493520000000000","last_tick":"1306493520000000000","span_ticks":"0"}